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A comparative investigation of compositional syntax and semantics in DALL·E and young children

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ABSTRACT

In this study we compared how well DALL·E 2 and DALL·E 3 visually represented the meaning of linguistic prompts also given to young children in comprehension tests. Sentences representing fundamental components of grammatical knowledge were selected from assessment tests used with several hundred English-speaking children aged 2–7 years for whom we had collected original item-level data. DALL·E 2 was given these prompts five times to generate 20 cartoons per item, for 9 adult judges to score. Results revealed no conditions in which DALL·E 2-generated images that matched the semantic accuracy of children, even at the youngest age (2 years). DALL·E 2 failed to assign the appropriate roles in reversible forms; it fared poorly on negation despite an easier contrastive prompt than the children received; it often assigned the adjective to the wrong noun; it ignored implicit agents in passives. As a follow-up study, we queried DALL·E 3 to generate 8 images each of the same prompts. This demonstrated a consistent failure to compositionally represent syntactic structure (in particular for negation and reversible prepositional phrases), albeit with some improvements from DALL·E 2. This work points to a clear absence of compositional sentence representations for both DALL·E 2 and DALL·E 3.

1. Introduction

Recent advances in artificial intelligence with deep learning have, for some researchers, bolstered the argument that exposure to enough positive data allows a general-purpose algorithm to develop language capacities that are effectively indistinguishable from those of a human. But while substantial attention has been placed on predicting sequences of words, there has been less research into how such sequences might map onto referents and events. In an exciting recent development, the text-to-image generator DALL·E 2 was celebrated for transforming natural language prompts into realistic images (Ramesh et al., 2022). DALL·E has often been described as “understanding” the natural language prompts it receive, in a manner that links the text to the world of referents (but see Baggio & Murphy, 2024). Yet, it is only very recently that researchers have called for more linguistically valid benchmarks for text-to-image models, that take into consideration *compositional* text prompts and involve “attribute bindings, object relationships, and logical reasoning” (Li et al., 2024). Here, we ask: Has DALL·E understood language to an extent comparable to young children?

2. Theories of child language acquisition

Theories of language acquisition in young children remain controversial. On one side are those who argue that language is just one of many skills to be learned, and that the size and complexity of the human brain makes it possible in a short amount of time because of the vast amount of input that surrounds children (Cantlon & Piantadosi, 2024). Children in the first 4 years of life hear approximately 12,000 words per day (Gilkerson et al., 2018). Most of these words are spoken in rich contexts by partners sympathetic to their level of understanding who adjust the complexity of their speech as the child shows evidence of understanding (Hirsh-Pasek et al., 2018; Snow, 1972). It is considerably more difficult to work out a precise measure of the number and complexity of *sentences* that the average child hears, but in the same study, Gilkerson et al. (2018) reported on average over 500 “conversational exchanges” per day to children aged 2–4 years. More recently, Bergelson et al. (2023) provided an estimate of 2.3 million adult utterances by the 3rd birthday, across a large sample of children in various countries. The input children receive is substantial. But the question remains: is it sufficient?

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In contrast, there are those who argue that the child goes well beyond the information given in the input, and is able to independently structure the heard speech into a grammar according to an innate, specialized device unique to the human brain (Yang et al., 2017). Adherents of this theoretical position have been able to show that children know not only what *can* be said, but more importantly what *cannot* be said; namely, constraints on grammatical form, which are not learnable from positive input (Leivada & Murphy, 2022). As Crain and Pietroski (2001: 161) observe:

[P]rojecting beyond experience is just one aspect of language acquisition. Children also fail to project beyond their experience in characteristic ways. It is this fact that most impresses nativists. The theoretical problem posed by human language learning is to explain why children project beyond their experience just so far and no further; the specific ‘angle’ of projection seems arbitrary (and idiosyncratic to linguistic projection).

In laying out the contrasts between the two major positions on language acquisition, we will characterize them as the *experience-dependent* account versus the *nativist* account. Contrasts between these positions are numerous: in particular, the putative learning mechanism, what can be explained, and the manner in which errors are corrected.

The experience-based account relies on a domain-general mechanism for learning, using statistical pattern recognition as the main procedure. It allows cut-and-paste formulaic learning (Tomasello, 2000), and analogy across structure treated at a surface level of similarity (Ambridge, 2020), avoiding theoretically-laden grammatical labels. Regular features will be learned rapidly as they produce common patterns, or construction types (Goldberg, 2019). But idiosyncratic features of language would also be easily learnable, if they are sufficiently frequent. In a language like English where idiosyncrasy abounds, this is a persuasive argument.

For error correction, a number of possibilities exist. First, conservatism – sometimes called entrenchment – the learner stays close to the information in the input and does not extend it without reason. Snyder (2011) has made a compelling case that children surprisingly rarely make errors of commission, e.g., saying *goed* or *foots*. Omission is the far more common form of error, in support of conservatism. Pre-emption is another option discussed, where a better form for the same function wins because of statistical frequency (Ambridge et al., 2008), possibly more useful than entrenchment (Samara et al. in press). Finally, advocates of the experiential view allow correction in the input to play a role. Even if there is no specific negative feedback on an error (Marcus, 1993), there may be enough failures of the interlocutor to understand a form, or caregiver’s repetitions of the child’s faulty utterances that in the process fix it, to alert the child to make a correction (Saxton et al., 2005). Finally, discourse and lexical/semantic properties are invoked as explanations of the inductions the child makes in syntax (e.g., Ambridge et al., 2008), though that is not unique to experientialists (e.g., Roeper & de Villiers, 1994; Thornton, 1995).

With respect to the nativist position, the learner comes equipped with a specific language-learning mechanism, namely Universal Grammar (UG). This provides a set of fundamental principles, and even some array of parametric options where languages vary, that dictate the course of learning. In recent accounts, the fundamentals are recursive, compositional Merge, along with proprietary representations that form the accessible objects for Merge, ultimately “lexical items” (Murphy, 2024b). The arguments adduced in favor of this mechanism revolve around the abstract nature of the grammar for which there is no possible negative evidence: constraints on interpretation of pronoun reference (Crain et al., 2017), or how quantifiers behave logically (Crain & Khlentzos, 2008), or how *wh*-questions can be formed (Crain et al., 2017; Roeper & de Villiers, 2011). Most generally, the *structure-dependence* of syntactic rules is, under this account, an innate bias on interpretation. Of course, counterproposals exist, drawing on semantics or discourse for guidance in acquisition (e.g., Kaiser et al., 2009).

However, the nativist argument is that there is nothing in the surface forms of language that can provide evidence of these from which to build such a grammar from the ground up. For many nativists such as ourselves, Merge is a fundamentally *unlearnable* computation, and there are no means to motivate a ‘gradualist’ account of it (e.g., ‘proto-Merge’ followed by ‘real’ Merge). The universality and simplicity of Merge speaks to its unlearnability. Processes such as *analogy across constructions* are demonstrated to be inadequate to capture the sophistication of children’s generalizations (Koring et al., 2020). Instead, these are part of the guiding program that comes with being human. Errors are explained by the child choosing wrong parametric settings at an early point in the learning process, but should not occur at all for deeper grammatical properties such as structure-dependence (Crain et al., 2017):

On this approach, all linguistic rules that child language learners formulate and execute are ones that analyse sentences into hierarchical structures. Children are precluded by human biology from treating sentences as strings. Therefore the biolinguistic approach predicts that children will not commit the kinds of errors that would result from the application of structure-independent operations. That is, children will never produce sentences using structure-independent operations, despite the simplicity of such operations and their ability to replicate the input children encounter.

For a possible counter-argument, see McCauley et al. (2021). Despite the strong role accorded to UG, others within this tradition argue that probabilistic learning is also needed for grammar (Leivada & Murphy, 2022; Yang, 2015), and offer general constraining principles to accommodate exceptions to rules.

However, the advocates of the experience-dependent position are also aligned with a set of arguments about the nature of the input that go well beyond the speech input itself. In early arguments against the Language Acquisition Device (LAD), proponents argued that the LAD neglected the role of careful tailoring by caregivers to the child’s needs. Initially the arguments had to do with properties such as “motherese” having clearer phrase boundaries, shorter sentences, reduced vocabulary, more melodic contours and crisper articulation (Snow, 1972). Later, it was claimed that human contact and emotional connection mediates the role of input: for example, that exposure to the radio wasn’t enough, nor was TV, without a human to converse with (Kuhl & Rivera-Gaxiola, 2008). More recently, the role of such properties as shared attention between mother and child to an event or object of significance to the child has gained increasing attention (Tomasello, 2008; Tomasello & Farrar, 1986), as has turn taking and conversational exchanges (Hirsh-Pasek et al., 2018). With the discussion changing to the context of learning grammar, it soon transpired that the reference of words, the meaning of sentences, and the intentions of interlocutors all came to the forefront.

2.1. The connection to AI models

The apparent extraordinary success of large language models (LLMs) is seemingly a benefit for the experience-dependent side – except that the input has none of the systematicity of guidance, meaningful contexts or intentionality that are supposedly needed to help a human child acquire language. Is it then just that the human child learns faster from less, because of caregiver guidance?

On the other side, nativist theorists find the results from LLMs unconvincing for the kind of specialized knowledge of grammar that humans have (Bolhuis et al., 2024; Katzir, 2023; Murphy & Piantadosi, 2023; Rawski & Baumont, 2023). However, superficially fluent AI output has persuaded many that LLMs have mastered language, or at least its formal and syntactic components (Hu et al., 2024). Some might ask, why bother considering the linguistic constraints, if we just never hear sentences that violate them? When is that knowledge ever needed? More significantly though, critics are talking not just about the sufficiency of the grammatical forms, but the fact that the discourse

produced by LLMs is often false, or “hallucinated”. On this side too then, reference and truth are central to the objections.

Consider also Kallini et al.’s (2024) assessments about artificial language model performance on ‘impossible’ human language structures. Kallini et al. (2024) show that GPT-2 exhibits a *preference* for natural languages – however, it crucially still performs very high on a range of impossible languages (as the authors put it, it does “not master our set of synthetic impossible languages as well”). Languages are either ‘possible’ or not – there is no continuum, they either allow structure-dependence of operations or they do not. It is a difference in kind, not degree. Importantly, in their Appendices Kallini et al. note that with respect to the core operation of *constituent identification* GPT-2 treated all of their various impossible languages the same, which in fact provides strong support for the generative argument about the limits of language models. In addition, word embeddings carry a decent amount of implicit information about language structure, and as such we should expect LLMs to be better at learning a specific language whose embeddings are learned from that.

Several recent approaches to child language from the input-side have claimed that early child language – usually before the age of three years – may be formulaic rather than rule-governed, and in particular, that a young child proceeds via a statistical analysis in terms of which words can combine into appropriate sentences (Lieven & Tomasello, 2008). Even structure-dependence itself has been explained away (Ambridge et al., 2008) by saying that children avoid structure-dependence errors because they are such expert distributional analyzers, a claim we consider highly dubious (see also Perkins & Lidz, 2021). The proponents of formulaic learning point to the limited kinds of sentences children produce, and the repetitive or even unanalyzed nature of early sentence frames. The limited system must eventually give way to more adult forms as the child learns to map speakers’ intentions across utterance types. This sets the stage for the possibility that text-to-image models like DALL-E will behave better than a young child, especially given the size of the training set. DALL-E needs no special motivation or caregiver guidance beyond sheer quantity of input.

In stark contrast, nativists argue that children have a rich adult-like grammar from the beginning, at least in terms of its intensional algorithms, elementary representations, and rules for mapping to conceptual systems, with the claim of ‘continuity’ taking center stage. That is, the grammars of children are claimed not to be structured differently from those of adults; there is continuity all the way down (Levy, 1983). Somewhat ironically, continuity is also espoused by experientialists, who argue that adults, like children, retain many formulaic and idiomatic structures.¹ For the nativist, the child is said to be acquiring grammar of an abstract, highly specific kind, not one resembling the outcome of statistical pattern learning. In fact, both Bencini and Valian (2008) and Yang (2016) demonstrate that the combinations children produce are far from formulaic, and in fact match those of adults in combinatorial diversity.

It is also common in research in the nativist tradition to take language comprehension as the indication of competence. However, the results are often covered by caveats: a certain amount of vocabulary is necessarily learned before children can handle sentences entailing that vocabulary, including which frames particular verbs allow; transitive, intransitive, dative, complement-taking. Sensitivity to frames, indicated by the accompanying noun phrase in certain positions vis-à-vis the verb, is strikingly early and can be used to identify the possible meaning of novel verbs (Fisher et al., 2020). On this account, given the special preparedness of the young child’s mind for human grammar of an abstract sort, it would be surprising if an LLM trained on statistical information could fare as well.

¹ Levy’s 1993 paper was called, “It’s frogs all the way down”. This invites the rival title, “It’s tadpoles all the way up”.

2.2. A further difference: bodies in a physical world

Consider another difference between AI models and humans – the role of embodiment in arriving at sentence interpretations (i.e., not the process of structure-formation itself, but rather the interpretive components). The fact that the child has a body and lives in a physical world may give them a radically different understanding of physical possibilities, intuitions and common sense than an AI. Even if one eschews the more radical premise of embodied cognition, there is a substantial literature on what Spelke and Kinzler (2007) called “core knowledge” systems; very early developed, potentially innate, understanding of fundamentals of balance, support, cause and effect, intention, occlusion, number, object permanence and even plant life. AI as manifest in DALL-E 2 has none of this, but the question is, does any of that matter for comprehending basic grammatical relations? Most experientialists would say answer affirmatively.

2.3. How can DALL-E and children be compared?

Recently, Leivada et al. (2023) demonstrated that DALL-E 2 falls short of the grammatical understanding of complex forms that linguists have argued are characteristic of human language; see also Marcus et al. (2022). However, to date no-one has directly compared DALL-E to young children. On the surface this would seem to present major problems: DALL-E produces a picture that illustrates a word, a phrase, or a sentence. Children are notoriously weak at drawing in the preschool years, even when their language capacity is rapidly becoming adult-like. We can, however, ask them to judge which of two or more pictures is the best representation of a sentence. DALL-E has to generate an appropriate match. Both child and AI must decide on a match between sentence meaning and image. The performance of DALL-E is considered to be a case of matching meaning to form, just like comprehension tests for children.

It is critical to ensure that this test is not too esoteric, as some phenomena can be theory-laden and contentious. The goal we set here is to choose very fundamental properties of language that every kind of theorist (generative grammar, construction grammar, etc.) would agree are central to the language user.

In this study, we focused on foundational aspects of grammar.

- 1) A basic operation required for **propositional semantics** is establishing the roles of agent and patient in a declarative sentence. This is tested by using reversible transitives for which the objects named by the noun could play either role with a chosen verb (“The car hit the truck”).
- 2) **Negation** is central to syntax-semantics (“One girl has shoes, one girl has no shoes”).
- 3) **Prepositions** can link nouns together, where the relation and order matter (“The book is on the cup”).
- 4) Embedded adjectives need to have appropriately **assigned scope**, e.g. ‘The kitten in a cup with a yellow ribbon’ does not mean that the cup or the kitten is yellow.
- 5) The **passive voice** entails more than reversed roles (“The cat is being dressed’ implies the existence of an external agent) (Roepers, 2004).

It is important to make the comparisons in behavior in very specific ways: using the same items to test, and large numbers of children from a representative sample. As there is only one DALL-E, the comparison will necessarily be with repeated elicitations of pictures to get a “sample” of DALL-E’s success.

To test DALL-E’s abilities, we compared how well children in comprehension tests understood the meaning of the same linguistic prompts we gave to DALL-E 2, to develop a comparable benchmark. To our knowledge this constitutes the first direct comparative test of DALL-E 2’s abilities and children’s abilities using the *same linguistic items*. Sentences were selected from assessment tests (Golinkoff et al.,

2017; Jackson et al., 2023; Seymour et al., 2005) used with several hundred English-speaking children aged 2–7 years for whom we had original item-level data (we refer the reader to these sources for more specific statistical details about performance).

Lastly, we conducted a follow-up study with DALL-E 3, in which we asked it to generate cartoon-style images of the same prompts, generating 8 images for each prompt. We assessed these images using the same criteria as above for DALL-E 2, and we provide all image outputs as appendices in Supplementary data.

3. Methods

We prompted DALL-E 2 with a sentence 5 times to generate 20 images each for an online survey for 9 adult judges (university educated). ‘Cartoon style’ was specified in the prompt to limit the possibility that DALL-E 2 could find online photographs with captions matching prompts. The 9 volunteer judges from different disciplines completed Google forms (560 images) to judge on a 3-point scale whether the images were (1) accurate depictions, (2) partially accurate but containing a misleading feature, or were (3) inaccurate (Fig. 1). The average score provided an adequacy percentage, such that, for example, an average of 3 would yield 0% adequacy, and an average of 2 would yield 50% adequacy, and an average of 1.5 would yield 75% adequacy. A second index considered only the scores of 3, and counted percentage correct. The percentage correct is the index most naturally compared to the children’s scores, but we considered both.

The searches for DALL-E 2 were run in May 2023. Several searches of similar commands were run to replicate the results (see Appendix B), crucially with no filtering or selection procedure. This method helps to extend the approach reported in Leivada et al. (2023), which did not solicit external, non-expert (hence, non-biased) judges for assessing image outputs.

The searches for DALL-E 3 were run in September 2024. For the DALL-E 3 results, given that this was a follow-up study of the direct comparison between DALL-E 2 and children, we calculated percentage correct, as above, and provide the accuracy scores across sentence types. We assigned DALL-E 3 outputs with accuracy values, but we provide a comprehensive output of all images generated (Appendix C).

Exact instructions provided to our judges are given in Appendix A. We also show extensive examples from negation, prepositional structures and transitive structures (Appendix B). Querying text-to-image applications in order to examine which properties of the text can be successfully captured and represented is an error-proof method that needs little defense, in particular given that we test the application in the domain that it has been tailored to perform best: transforming natural language prompts into realistic images.

Data from children across various ages were taken from sources providing large behavioral samples for performance on each prompt. These were mostly picture choice tasks from large language assessments samples, e.g. QUILS-TOD (Jackson et al., 2023), QUILS (Golinkoff et al., 2017), DELV (Seymour et al., 2005), with the exception of one act-out task (Chan et al., 2010). Percentage correct was taken as the comparable score for children. Importantly, while children had to select from multiple pictured options, DALL-E had to create a novel image, generating a divergence in typical task demands and processing strategies, yet we argue that the underlying demand for a compositional meaning was salient in both tasks. Given that it is not feasible to ask 2-3-year-old children to draw referents for even simple linguistic structures, this was an unavoidable difference. Due to the potentially misleading nature of direct statistical comparisons between divergent tasks, we report here grouped averages (Figs. 5–9), à la Leivada et al. (2023).

4. Results

4.1. DALL-E 2 assessment

Results revealed no conditions in which DALL-E 2-generated images that matched the semantic accuracy of children, even at the youngest age (2 years) (Fig. 4). DALL-E 2 failed to assign the appropriate roles in reversible transitive and prepositional phrase forms; it fared poorly on negation despite an easier contrastive prompt² than the children received (Fig. 2); it often assigned the adjective to the wrong noun; it ignored implicit agents in passives (Figs. 2 and 3). Across the board, even children as young as 2 years-old outperformed DALL-E 2 (Figs. 5–10).

4.2. DALL-E 3 assessment

To follow up on our direct comparison between young children and DALL-E 2 (given that, above, we have directly established such a comparison), we queried the latest model, DALL-E 3, in September 2024. We asked it to generate, as above, cartoon-style images of the same stimuli prompts. DALL-E 3 has a limit of outputting 2 images at once, so we prompted it 4 times, leading to a generation of 8 images each for all stimuli. We provide the full outputted images as appendices (Appendix C). Below, we provide some example outputs, and the percentage correct across the categories (Figs. 11–12) (note that for the passive construction we had to use “The cat is being dressed”, since the prompt we used above for DALL-E 2, “The cat is being undressed”, was considered by the latest version of DALL-E to be problematic).

DALL-E 3 performed poorly on negation, reversible PPs and advanced PPs. For some constructions we found 0% accuracy (e.g., “The ball is behind the dog”, “The umbrella is below the swing”). For PPs with adjectives, reversible transitives and passives, it performed better than DALL-E 2 and between children ages 3–6 across these construction types.

5. Discussion

This study provides the first direct comparison between child language comprehension and artificial text-to-image models. Our work points towards an absence of compositional sentence representations for DALL-E 2 and 3, both of which made numerous errors. The most serious fell into three categories that caused the participants to judge the depiction as a failure.

- Failure to distinguish the actors in the scene so that one could not tell which was which. If we cannot tell which one is the horse and which one is the elephant, how are we to judge the depicted event as “The bear is pushing the lion”?
- Reversal of roles: the agent/patient were reversed, ignoring fundamentals of grammar, e.g., “The sheep is kicking the cat” in which the cat is kicking the sheep.
- The action or relationship was so strange it could not be evaluated as a satisfactory depiction, e.g., “The firefighters are between the chairs”.

Error types (a) and (b) are clear failures to use the appropriate grammar, though they differ. Type (a) seems more like a “cheat”, where (b) clearly ignores semantic roles. Type (c) reflects the lack of world knowledge of how things can be located with respect to each other. These types were not uniformly distributed across the examples, but

² Children were asked to choose from a set of boys: “Find the boy with no shoes”. We asked DALL-E 2 to depict: “One boy has shoes, one boy has no shoes”. This was necessary to test whether the negation was being attended to at all.



“The umbrella is below the swing”

Do you think this image is

- (1) accurate
- (2) includes a potentially misleading feature
- (3) inaccurate

Explain your reason below:



“The kitten is in a cup with a yellow ribbon”

Do you think this image is

- (1) accurate
- (2) includes a potentially misleading feature
- (3) inaccurate

Explain your reason below:

Fig. 1. Example questions for the adult judges (Source: DALL-E 2).

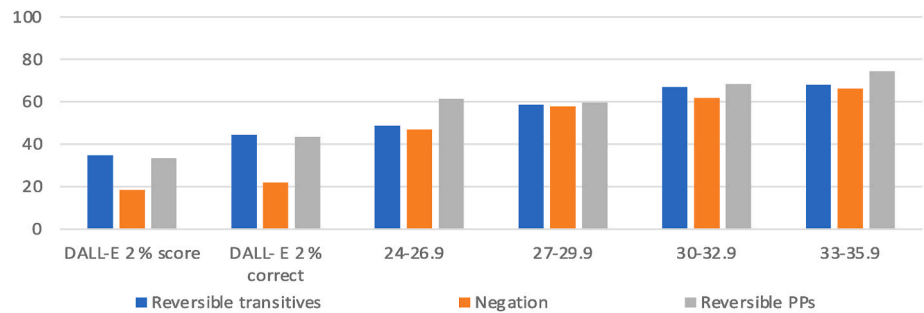


Fig. 2. Average performance on three early forms. The performance of DALL-E 2 is compared to children aged 24-35 months-old (age is in months). Percentage score = adequacy metric; percentage correct = accuracy metric (see Methods). Y-axis = percentage correct (for examples, see Appendix B).

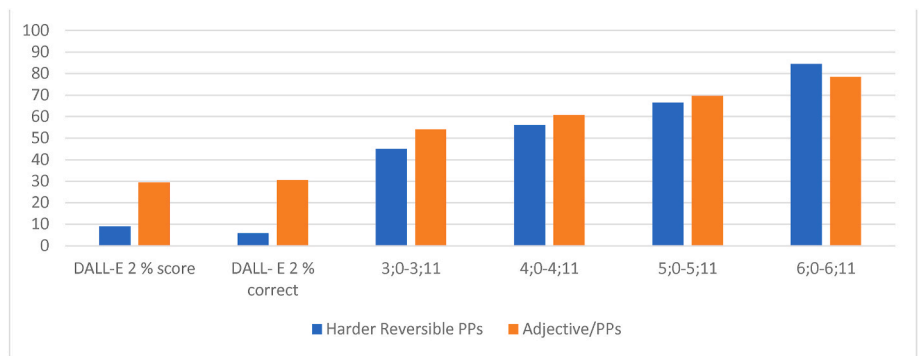


Fig. 3. The performance of DALL-E 2 compared to children aged 3–6 years (age is in years; months). Y-axis = percentage correct.

type (b) occurred with all examples but one. Clearly, children’s expertise in core knowledge of objects seems to contribute. For example, consider the example (Fig. 4) of “The firefighters are between the chairs”. DALL-E 2 in this and other cases did not respect ordinary assumptions about human bodies in space, but instead wove the firefighters and chairs

together. DALL-E 3 also made a number of spatial errors for the meaning of ‘between’. This kind of error goes beyond adding extra digits to hands, or the statistical bias of representing women rather than men holding babies, and represents a fundamental error in compositional syntax-semantics. In addition, it may also represent a lack of clear semantic

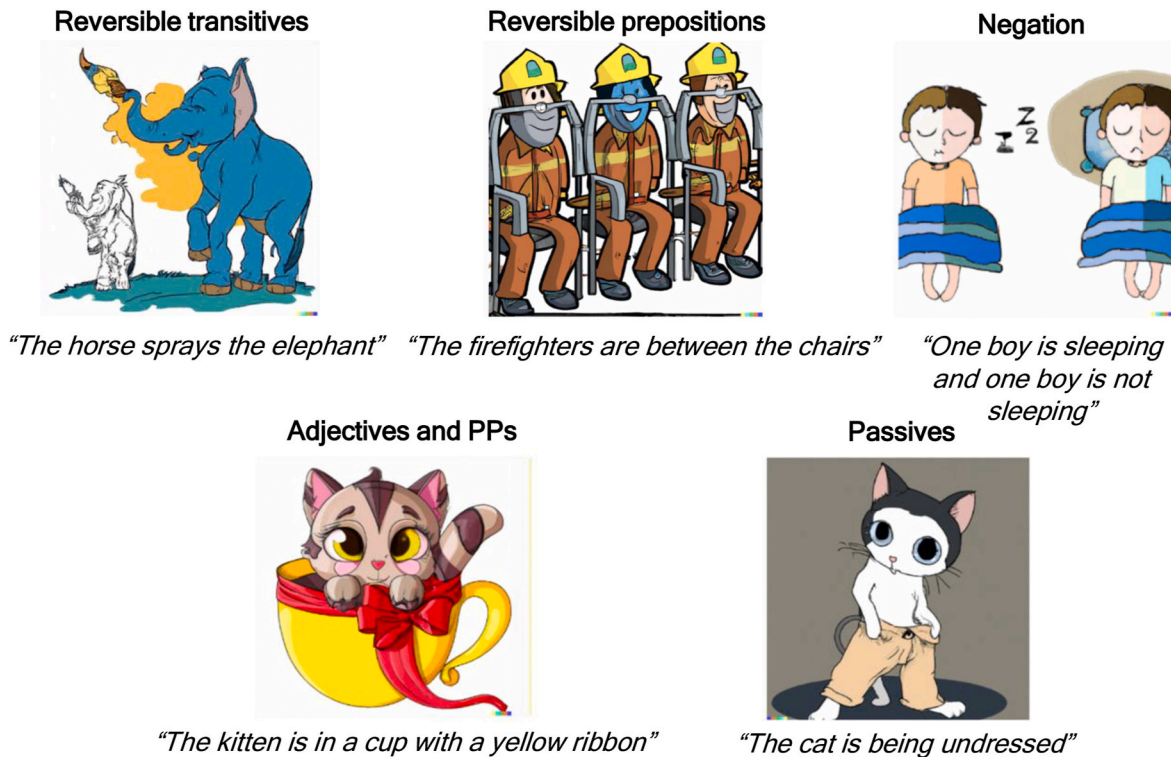


Fig. 4. The performance of DALL-E 2 on a range of compositional syntactic processes (see Appendix B) (Source: DALL-E 2).

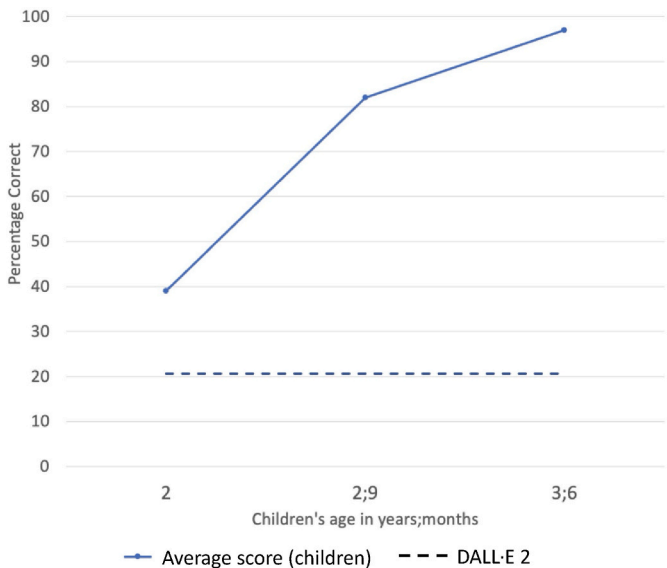


Fig. 5. The performance of DALL-E 2 vs. children tasked with acting out the sentence meaning of reversible transitive sentences. We note that Chan et al. (2010) provided items but not item-level data, and so we averaged for comparison.

and pragmatic rules of inference, with some recent researchers arguing that artificial language models are currently unable to achieve pragmatic reasoning competence (Mahowald et al., 2024). Human children are able to construct a grammar with direct links to compositional meaning, connecting language to internal cognitive models of the world – an architecture that seems absent in DALL-E, which does not seem able to reliably and robustly construct linguistically coherent representations. Importantly, even for DALL-E 3, while the level of graphical fidelity has surely improved, the degree of representational stability

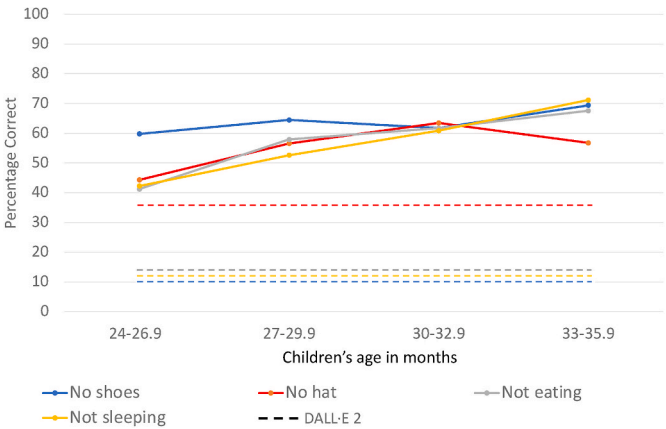


Fig. 6. The performance of DALL-E 2 vs. children on negatives. Dotted lines represent DALL-E 2 performance on correspondingly colored structures.

remains an issue, i.e., DALL-E 3 performed at around 50% accuracy for many of our prompts involving two agents (X does P and Y does Q), indicating no robust compositional representation. The most salient improvement from DALL-E 2 to DALL-E 3 was its ability to represent agents (e.g., with prompts involving “dressed” and “bumps”).

What are the implications of these findings for theories of language acquisition? In recent work, visual and auditory input was recorded from a helmet with a camera on a single human child from 6 to 25 months of age (Vong et al., 2024). 61 h of data were fed to a “multimodal AI” to determine if the system could learn the referents of words captured on audio, from the visual input. Despite the great variability in the likelihood of objects being physically present when discussed, the machine learning model apparently succeeded 62% of the time in recognizing referents in a subsequent test, compared to 25% by an untrained AI. This is a start, but it is very far from grammar and event perception.

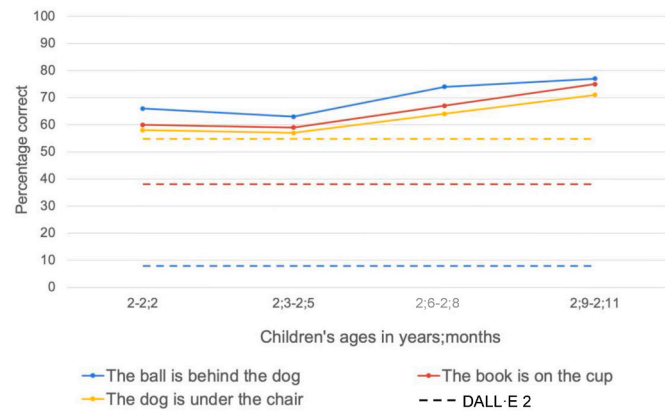


Fig. 7. The performance of DALL-E 2 vs. children on reversible prepositions. Dotted lines represent DALL-E 2 performance on correspondingly colored items.

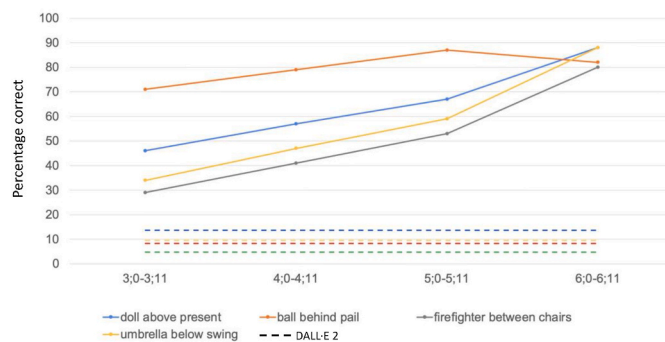


Fig. 8. The performance of DALL-E 2 vs. children on complex prepositional phrases. Dotted lines represent DALL-E 2 performance on correspondingly colored items.

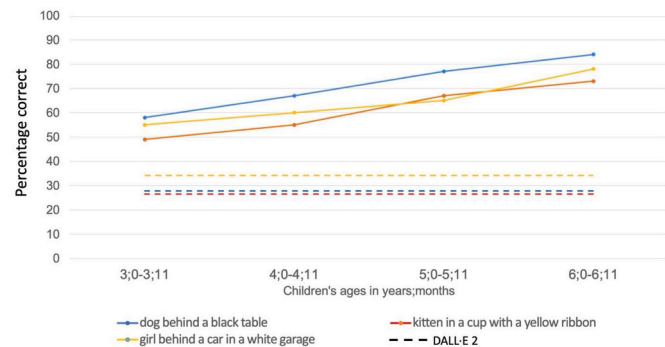


Fig. 9. The performance of DALL-E 2 vs. children on prepositional phrases with adjectives. Dotted lines represent DALL-E 2 performance on correspondingly colored items.

Those theorists who are advocates of experiential, shallow, probabilistic learning might argue that DALL-E needs a more nuanced training set that builds and guides like a human caregiver does for a child, rather than prioritizing the sheer quantity of input. Most experientialists also concede a degree of innateness in humans, but in the domain of social-cognitive skills, not language per se. However, the device must be able to generalize beyond trained examples, and to do that, analogy is an insufficient procedure. As [Crain et al. \(2017\)](#) point out, the child does not make simple inductions, but instead, abstract ones that often cross construction types. We need to account for the ways that children go beyond the information given. The lines of “stimulus generalization” may be given in advance.

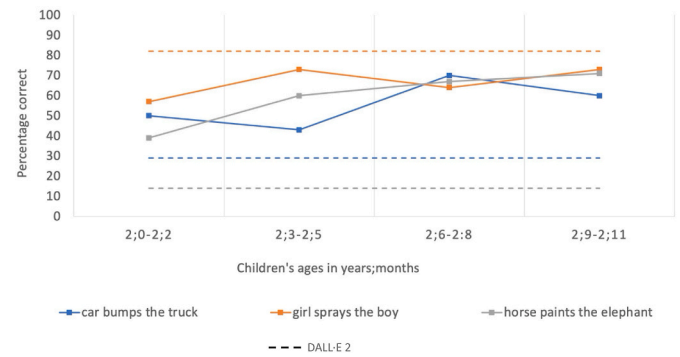


Fig. 10. The performance of DALL-E 2 vs. children on reversible transitives. Dotted lines represent DALL-E 2 performance on correspondingly colored items.

We argue that the data are expected for a learning device that lacks basic competence in compositionality. We confirm and extend the findings from [Leivada et al. \(2023\)](#) that this fundamental aspect of human language is lacking from DALL-E 2 (see [Murphy, 2023](#); [Murphy & Leivada, 2022](#); [Wasielewski, 2023](#)). This feature is present early and robustly in human children.

One avenue for future research that could overcome some of the limitations of our survey would be to give an LLM the outputs of DALL-E 2/3 and query that LLM to perform a classification/recognition task that is identical to the one given to the children in our reported database. [Li et al. \(2024\)](#) report DALL-E 3, Pika, Gen2 and Stable Diffusion failures with sentences involving object differentiation and quantification (e.g., “Every lamp in the street is lit, except for one that remains dark”). Given that our study attempted to directly link text-to-image generator performance with children’s performance, we were constrained in our stimuli complexity, and future work should explore more carefully the representation of complex syntactic phenomena.

Our results also dovetail into recent concerns about the lack of grammatical competence for language models, and the presence of certain response biases that deviate from standard assumptions concerning common processing strategies that humans possess for language ([Dentella et al. 2023a,b](#)). The importance of syntax-specific inductive biases cannot be underestimated ([Sartran et al., 2022](#)), in particular given that grammar reflects the organization of a specific mode of thought that appears to be unique to humans ([Hinzen, 2006](#); [McCarty et al., 2023](#); [Murphy, 2015, 2020, 2024a](#); [Murphy et al., 2022](#)). We also note the promising direction that neurosymbolic approaches are offering the field ([Murphy, 2024c](#); [Trinh et al., 2024](#)), which seem ripe for adaptation towards syntactic representations given their moves towards cognitively plausible architectures ([Marcolli, Chomsky, & Berwick, 2023](#); [Murphy, Holmes, & Friston, 2024](#)) constituting a hybrid of rich prior knowledge and sophisticated reasoning techniques ([Marcus, 2020](#)). In addition to grammar, the pragmatics (and ‘super pragmatics’; [Schlöder & Altshuler, 2023](#)) of language processing also appears absent in current artificial models. Future research should explore additional aspects of functional grammatical structure to test these limits further ([Sonkar et al., 2022](#)), given the centrality of function words in regulating syntactic information (for broader discussion, see [Merrill et al., 2024](#)). Lastly, future work could seek to approximate the type of task conducted by both DALL-E 2/3 and child participants, to make the task demands as similar as possible, given that in our comparison children had to select from multiple options whereas DALL-E had to create a novel image (for related discussion of the problems of defining ‘task similarity’, see [Leivada et al., 2024](#)). Nevertheless, despite divergent task demands, the stark accuracy differences reported here across all of our linguistic structures for DALL-E 2, and most of the structures for DALL-E 3, provides strong evidence for the absence of a higher-order compositional apparatus for DALL-E, which forms the fundamental basis of human language ([Partee, 1984](#); [Pietroski, 2018](#)).

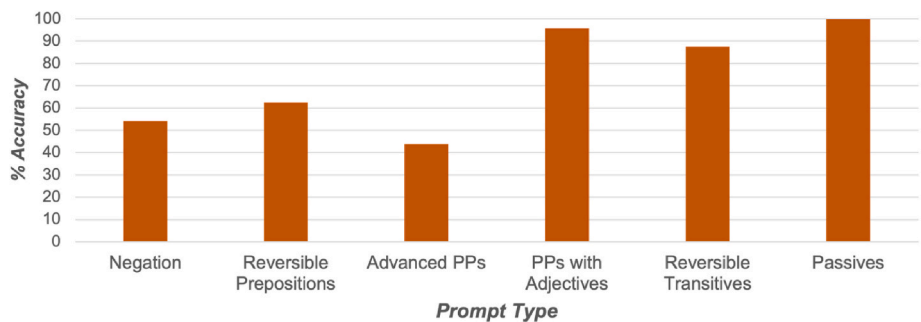


Fig. 11. The accuracy of DALL-E 3 across a range of compositional syntactic processes (see Appendix C).

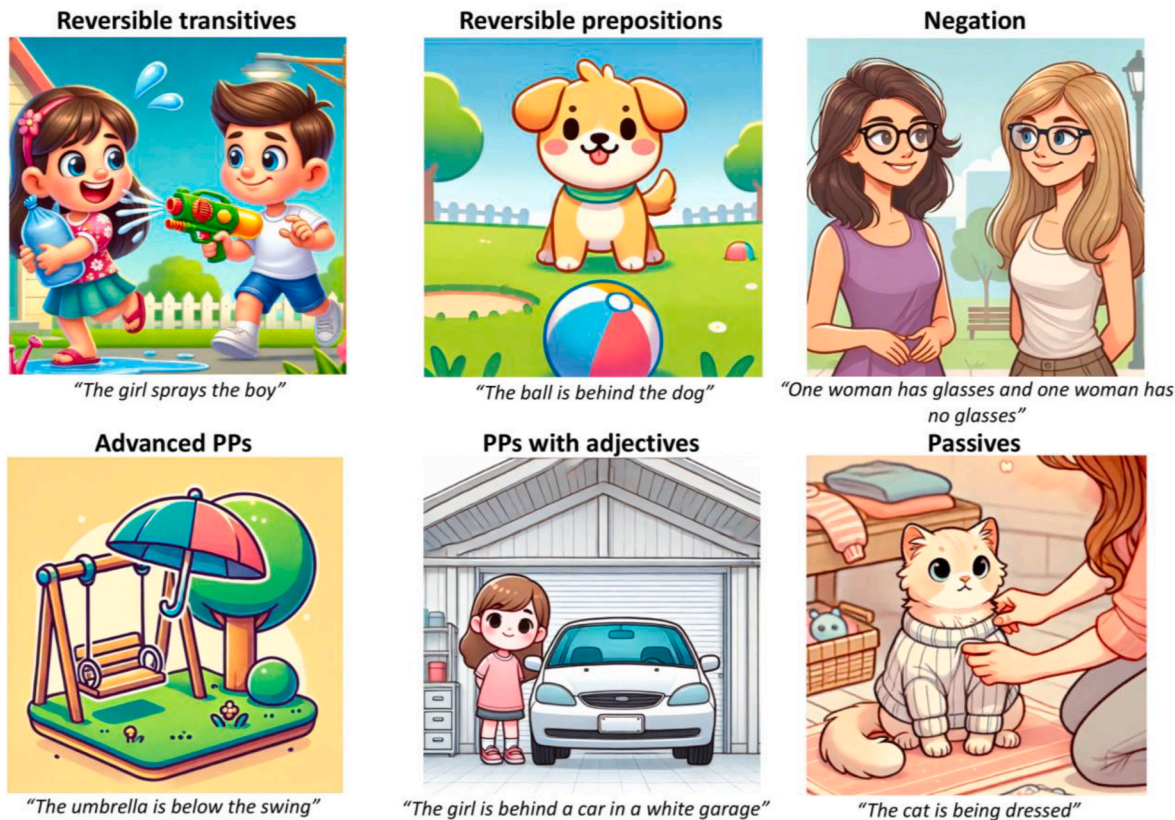


Fig. 12. Examples of DALL-E 3 performance on a range of compositional syntactic processes (see Appendix C) (Source: DALL-E 3).

CRediT authorship contribution statement

Elliot Murphy: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jill de Villiers:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization. **Sofia Lucero Morales:** Visualization, Software, Methodology, Investigation, Formal analysis, Data curation.

Data availability statement

All data used is provided in the appendices.

Declaration of the use of AI

The authors made no use of generative AI in the writing, analysis or

presentation of this manuscript.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ssaho.2025.101332>.

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